Tide retrospective: the nondegenerate λ-package (unequal-base programme, stages B3–B5)

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

2026-08-10

1 Setting

The closing tide of programme B′. Base recovery (previous tide) removes the equal-base hypothesis; what remains is composition and packaging, which the scoping consult predicted would need "no modification of the pairwise machinery". It didn't.

2 The thing formalised

Five declarations in `Laplace/OneD/LambdaPackage.lean`, sorry-free with zero warnings (gate verified via import and `.olean`). The t-to-q data transfer, factored from the recovery theorem's proof into a reusable helper; the variable-base recovery[1] (recover the base from second moments, substitute, apply the comparison theorem verbatim); the $k = 1$ package in the note's normalisation[2]:

> For two potentials $(\lambda_\nu/2)\,x^2 + \sum_{r=1}^{R} c_r^\nu x^{2+r}$ with uniformly positive profiles, eventually-equal Gibbs moments at x^2 and at each x^{2+r} force $\lambda_1 = \lambda_2$ and $c^1 = c^2$;

and the opening move of germbij Theorem 3.1 in its native form,

$$t\,\langle x^2\rangle_t \;\longrightarrow\; 1/\lambda$$

(`gibbs_secondMoment_rate`), with the $k = 1$ constant $M_2(\lambda/2) = 1/\lambda$ evaluated through $\Gamma(3/2) = \frac{1}{2}\Gamma(1/2)$.

With this, programme B′ is complete, and so is the *polynomial algebraic core* of Theorem 3.1: every coefficient of a finite nondegenerate jet is pinned by moment asymptotics. Per the scoping consult's explicit warning, this is not yet Theorem 3.1 itself — the smooth-loss reduction (programme C: an asymptotically stable recovery interface, localisation, Taylor-remainder transfer) is the remaining, genuinely analytic gap.

3 Roadblocks and resolutions

Small and catalogued: `Real.sqrt_mul_self` is $\sqrt{a \cdot a} = a$ while the needed identity $\sqrt{a} \cdot \sqrt{a} = a$ is `Real.mul_self_sqrt`; the radical-atom `set` pattern applied verbatim twice; and unreduced literal exponents (`(st`$^{-1}$`)^(2*1)`) block `field_simp` until `pow_mul`/`pow_one` normalise them.

[1] `polynomialJet_recovery_variableBase`.

[2] `nondegenerateJet_recovery`.

A process near-miss worth recording: the first draft of this tide's numerical-check section was written in the same shell command as the check itself — before the output existed — and carried invented values. It was caught and corrected against the actual output before any commit. The rule is now sharpened in the tide log: the check runs in one command; the log text is written in a later command, quoting.

4 The deliberation and check

No fresh consult (the scoping consult specified B3–B5 with proof structures; carried votes recorded in the log). The numerical check: $t\,\mathbb{E}_t[x^2]$ for $\lambda = 1.7$ with cubic and quartic perturbations reaches 0.587314 at $t = 800$ against $1/\lambda = 0.588235$, and $\Gamma(3/2)/\Gamma(1/2) = 0.5$.

5 What the next programme inherits

Programme C, scoped in the same consult: the asymptotically stable recovery theorem (the existing proofs' contradictions only need the observed differences to vanish at the coefficient-sensitive scale — the `_of_tendsto` interfaces built along the way are exactly this shape), then quadratic control and domain splitting, rescaled Taylor remainders, cutoff transfer, and the full-jet quantifier packaging. That programme is the note's "Justification" paragraph made formal, and it is where the germbij Theorem 3.1 marker will eventually land.